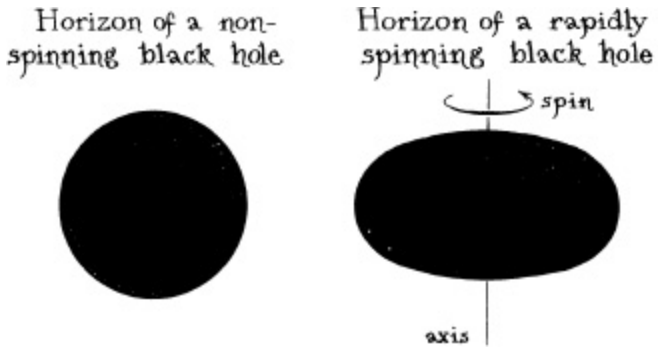



**7.9 The shapes of the horizons of two black holes, one (*left)* not spinning, and the other *(right)* spinning with a spin rate 58 percent of the maximum. The effect of the spin on the horizon shape was discovered in 1973 by Larry Smarr, a student at Stanford University who was inspired by Wheeler.**

For a hole with the mass of the Sun, the maximum spin rate is one revolution each 0.000062 second (62 microseconds). Since the hole's circumference is about 18.5 kilometers, this corresponds to a spin speed of about (18.5 kilometers)/(0.000062 second), which is about the speed of light, 299,792 kilometers per second (not entirely a coincidence!). A hole whose mass is 1 million Suns has a 1 million times larger circumference than a l-solar-mass hole, so its maximum spin rate (the rate which makes it spin at about the speed of light) is 1 million times smaller, one revolution each 62 seconds.

In 1969, Roger Penrose (about whom we shall learn much in Chapter 13) made a marvelous discovery. By manipulating the equations of Kerr's solution to the Einstein field equation, he discovered that a spinning black hole stores *rotational energy* in the swirl of space around itself, and because the swirl's energy is *outside* the hole's horizon and not inside, this energy can actually be extracted and used to power things. Penrose's discovery was marvelous because (as Demetrios Christodoulou later showed) the hole's rotational energy is huge. If the hole spins at its maximum possible rate, its efficiency at storing and releasing energy is 48 times higher than the efficiency of all the Sun's nuclear fuel. If it were to burn all its nuclear fuel (actually, it will not burn *all),* the Sun would only be able to convert a fraction 0.006 of its mass into heat and light. If one were to extract all of a fast-spinning hole's rotational energy (thereby halting its spin), one would get out 48 × 0.006 = 29 percent of the hole's mass as usable energy.